Stochastic Calculus

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Yao Xiao

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1 Construction of Brownian Motion

1.1 Brownian Motion and Its Motivation

Loosely speaking, Brownian motion is a "random path" that satisfies the following properties.

- The path is continuous.

- At any given point in time, where the path goes next is *independent* of where the path came from.
- At any given point in time, where the path ends up in any finite time is normally distributed.

1.2 Symmetric Random Walk

It is always easier to start with a discrete case. Consider a symmetric random walk (SRW). Let X_j's be independent and identically distributed such that $\mathbb{P}[X_j = 1] = \mathbb{P}[X_j = -1] = 1/2$. We now define $M(n)$ such that

$$M(0) = 0, \qquad M(n) = \sum_{j=1}^{n} X_j. \tag{1}$$

The process $M(n)$ is called a symmetric random walk. With each step, it either steps up or steps down one unit, and each of the two possible moves is equally likely.

1.2.1 Increments of the SRW

SRW has independent increments, which means that if we choose nonnegative integers $0 = n_0 < n_1 < \cdots < n_k$, then the random variables

$$M(n_1) = M(n_1) - M(n_0),\ M(n_2) - M(n_1),\ \cdots,\ M(n_k) - M(n_{k-1}) \tag{2}$$

are independent. Each of these random variables

$$M(n_{i+1}) - M(n_i) = \sum_{j=n_i+1}^{n_{i+1}} X_j \tag{3}$$

is called an increment of the random walk. It is the change in the position of the random walk between times n_i and n_{i+1}. Increments over non-overlapping time intervals are indeed independent, since X_j's are independent. Moreover,

$$\mathbb{E}[M(n_{i+1}) - M(n_i)] = \sum_{j=n_i+1}^{n_{i+1}} \mathbb{E}[X_j] = 0, \tag{4}$$

and

$$\mathrm{Var}[M(n_{i+1}) - M(n_i)] = \sum_{j=n_i+1}^{n_{i+1}} \mathrm{Var}[X_j] = n_{i+1} - n_i, \tag{5}$$

since $\mathrm{Var}[X_j] = 1$ and this holds by independence.

1.2.2 Martingale Property of the SRW

In order to define martingale we need the notion of σ-algebra. There is a very important, nontechnical reason to include σ-algebras in the study of stochastic processes, and that is to keep track of the amount of information. The temporal feature of a stochastic process suggests a flow of time, in which at each moment $t \geq 0$, we can talk about the past, the present, and the future, and can ask about how much an observer of the process knows about it at present, as compared to how much he know at some point in the past or will know at some time in the future.

Definition 1.1. A *filtration* on $(\Omega, \mathscr{F})$is a family $\{\mathscr{F}_t\}_{t\geq 0}$ of σ-algebras $\mathscr{F}_t \subseteq \mathscr{F}$, such that

$$0 \leq s \leq t \implies \mathscr{F}_s \subseteq \mathscr{F}_t, \tag{6}$$

i.e., $\{\mathscr{F}_t\}_{t\geq 0}$ is increasing.

Thus informally, one can think about the σ-algebra $\mathscr{F}_t$ in the above definition as the "knowledge" available at time t. The fact that $\{\mathscr{F}_t\}_{t\geq 0}$ is monotonically increasing just reflects the fact that as time goes on, the amount of available information cannot decrease.

Definition 1.2. A stochastic process $\{Z_t\}_{t\geq 0}$ is called a *martingale* with respect to a filtration $\{\mathscr{F}_T\}_{t\geq 0}$ if

(1) Z_t is $\mathscr{F}_t$-measurable for all t.

(2) $\mathbb{E}[|Z_t|] < \infty$ for all t.

(3) $\mathbb{E}[Z_t|\mathscr{F}_s] = Z_s$ for all $s \leq t$.

The first two conditions are for technical reasons, since we can only define measures on measurable sets, and we assume that the first moments are finite. The most important property in the above definition is the third property. Simply put, it states that the "best guess" for X_t given information available at moment s is X_s, and thus the process X_t has neither upward or downward drift.

Now we want to check that the SRW is a martingale. First, we need to specify a filtration $\{\mathscr{F}_t\}_{t\geq 0}$. Let $\mathscr{F}_n$ be the σ-algebra (read information) generated by the set of random variables $X_1, X_2, \cdots, X_n$.

(1) $M(n)$ is $\mathscr{F}_n$-measurable since it depends only on X_j, $j \leq n$, *i.e.*, on the information available at time n.

(2) From the fact that $|M(t)| \leq t$, we can easily see that $\mathbb{E}[|M(t)|] \leq t < \infty$.

(3) Using the fact that $\mathbb{E}[M(s)|\mathscr{F}_s] = M(s)$ and the fact that the random variable $M(t) - M(s)$ is independent of σ-algebra $\mathscr{F}_s$, we can deduce that

$$\begin{aligned}\mathbb{E}[M(t)|\mathscr{F}_s] &= \mathbb{E}[M(t) - M(s) + M(s)|\mathscr{F}_s] = \mathbb{E}[M(t) - M(s)|\mathscr{F}_s] + \mathbb{E}[M(s)|\mathscr{F}_s] \\ &= \mathbb{E}[M(t) - M(s)] + M(s) = 0 + M(s) = M(s). \end{aligned} \tag{7}$$

1.2.3 Quadratic Variation of the SRW

The quadratic variation of a discrete stochastic process up to time t is defined as

$$\langle M, M\rangle_t = \sum_{j=1}^{t} (M(j) - M(j-1))^2. \tag{8}$$

The quadratic variation up to time t along the path is computed by taking all the one-step increments $M(j)-M(j-1)$ along that path, squaring these increments and then summing them up. Clearly, the SRM increments can only take values ± 1, and thus $\langle M, M\rangle_t = t$. Note that in general, the quadratic variation of a stochastic process is stochastic itself. The fact that the quadratic variation in this case is deterministic is a special feature of the SRW.

1.3 Scaled Symmetric Random Walk

To approximate a Brownian motion, we speed up time and scale down the step size of an SRW. More precisely, we fix a positive integer n and define the scaled SRW at rational points k/n as

$$B^{(n)}\left(\frac{k}{n}\right) = \frac{1}{\sqrt{n}} M(k). \tag{9}$$

At all other points we define $B^{(n)}(t)$ as the linear interpolation between its values at the nearest points of the form k/n. The following properties of the scaled SRW could be easily proved using the corresponding properties of the SRW and we leave their proof as an exercise.

(1) Independence of increments, *i.e.*, for all rational numbers $0 = t_0 < t_1 < \cdots < t_n$ of the form k/n, the random variables

$$B^{(n)}(t_1) - B^{(n)}(t_0),\ B^{(n)}(t_2) - B^{(n)}(t_1),\ \cdots,\ B^{(n)}(t_n) - B^{(n)}(t_{n-1}) \tag{10}$$

are independent.

(2) $\mathbb{E}[B^{(n)}(t) - B^{(n)}(s)] = 0$, $\mathrm{Var}[B^{(n)}(t) - B^{(n)}(s)] = t - s$.

(3) $\mathbb{E}[B^{(n)}|\mathscr{F}_s] = B^{(n)}(s)$.

(4) The quadratic variation of the scaled SRW $\left\langle B^{(n)}, B^{(n)}\right\rangle_t = t$.

1.3.1 Limiting Distribution of the Scaled SRW

Now the idea is to let $n \to \infty$ in the scaled SRW and in the limit we obtain something satisfying the intuitive properties outlined for Brownian motion. By definition,

$$B^{(n)}(t) = \frac{1}{\sqrt{n}} M_{nt}. \tag{11}$$

Since $M_k = \sum_{j=1}^{k} X_j$ has a binomial distribution with parameters k and $1/2$, we can easily calculate the distribution of $B^{(n)}(t)$. In particular, one can draw the histogram of $B^{(n)}(t)$. We know that the random variable $B^{(n)}(t)$ has mean 0 and variance t. If we draw on top of the histogram of $B^{(n)}(t)$ the graph of normal density with mean 0 and variance t, we will see that the distribution of $B^{(n)}(t)$ is nearly normal. In fact, the central limit theorem asserts that $B^{(n)}(t)$ converges *in distribution* to the normal random variable $\mathscr{N}(0,t)$, but let us not use this theorem and prove the convergence of $b^{(n)}(t)$ from scratch. The whole point of this exercise is to learn a very powerful mathematical tool known as *characteristic functions*.

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Definition 1.3. Let X be a random variable with distribution $\mathbb{P}$. The *characteristic function* of X is defined as

$$\phi_X(u) = \mathbb{E}[e^{iuX}] = \int e^{iux} d\mathbb{P}[x] \tag{12}$$

for every $u \in \mathbb{R}$.

Example 1.4. (1) If $X = \mathrm{Ber}(p)$, then

$$\phi_X(u) = p(e^{iu} - 1) + 1. \tag{13}$$

(2) If $X = \mathrm{Bin}(n,p)$, then $\mathbb{P}[X = k] = \binom{n}{k} p^k (1-p)^{n-k}$, and

$$\phi_X(u) = (p(e^{iu} - 1) + 1)^n. \tag{14}$$

(3) If $X = \mathrm{Pois}(\lambda)$, then $\mathbb{P}[X = k] = e^{-\lambda}\lambda^k/k!$, and

$$\phi_X(u) = \sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} e^{iuk} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{iu})^k}{k!} = \exp\left(\lambda(e^{it} - 1)\right). \tag{15}$$

(4) If X is a Gaussian random variable $\mathscr{N}(m, \sigma^2)$, then

$$\phi_X(u) = \exp\left(ium - \frac{u^2\sigma^2}{2}\right). \tag{16}$$

The importance of the characteristics functions could be summarized in the following two theorems.

Theorem 1.5. If two random variables have the same characteristic function, then they have the same distribution. Of course, the converse also holds, in that if two random variables have the same distribution, then they have the same characteristic function.

Theorem 1.6 (Paul Lévy theorem)**.** Let X_n be a sequence of random variables on $\mathbb{R}$. Then

(1) If X_n converges to X in distribution, then $\phi_{X_n}(u) \to \phi_X(u)$ as $n \to \infty$ for every u, and in fact this convergence is uniform.

(2) If $\phi_{X_n}(u) \to \phi_X(u)$ for every u pointwise, then X_n converges to X in distribution.

(3) If $\phi_{X_n}(u)$ converges to some function $\phi(u)$ for every u pointwise, and $\phi(u)$ is continuous at 0, then there exists a unique random variable X, such that ϕ is the characteristic function of X and X_n converges to X in distribution.

We use Part (2) of Paul Lévy theorem to prove that $B^{(n)}(t)$ converges to the normal distribution with expectation 0 and variance t. We have that

$$\phi_{B^{(n)}(t)}(u) = \mathbb{E}\left[\exp\left(iuB^{(n)}(t)\right)\right] = \mathbb{E}\left[\exp\left(\frac{iuM_{nt}}{\sqrt{n}}\right)\right] = \mathbb{E}\left[\exp\left(\frac{iu}{\sqrt{n}}\sum_{j=1}^{nt} X_j\right)\right]$$
$$= \left(\mathbb{E}\left[\exp\left(\frac{iuX_j}{\sqrt{n}}\right)\right]\right)^{nt} = \left(\frac{1}{2}\exp\left(\frac{iu}{\sqrt{n}}\right) + \frac{1}{2}\exp\left(\frac{-iu}{\sqrt{n}}\right)\right)^{nt}. \tag{17}$$

Now we need to show that as $n \to \infty$, the above result converges to the characteristic function of the normal random variable with mean 0 and variance t, *i.e.*, to $e^{-u^2t/2}$. Expanding in Taylor series, we can see that

$$\frac{1}{2}\exp\left(\frac{iu}{\sqrt{n}}\right) + \frac{1}{2}\exp\left(\frac{-iu}{\sqrt{n}}\right) = 1 - \frac{u^2}{2n} + O(n^{-3/2}), \tag{18}$$

and thus

$$\phi_n(u) = \left(1 - \frac{u^2}{2n} + O(n^{-3/2})\right)^{nt} \to \exp\left(-\frac{u^2}{2n}\cdot nt\right) = \exp\left(-\frac{u^2t}{2}\right). \tag{19}$$

Therefore, we can conclude that $B^{(n)}(t)$ converges in distribution to a normal variable with expectation 0 and variance t. Strictly speaking, we have just demonstrated that for a fixed moment t, the distribution of $B^{(n)}(t)$ converges to $\mathscr{N}(0,t)$, but we did not prove the convergence of the whole stochastic process to Brownian motion. To do this, one has to further prove the convergence of the finite dimensional distributions and use *Prokhorov's theorem*.

1.4 Brownian Motion: A Formal Definition

A one-dimensional Brownian motion is a continuous-time stochastic process that satisfies the following properties.

- **Independence of increments:** If $t_0 < t_1 < \cdots < t_n$, then the random variables
$$B(t_0), B(t_1) - B(t_0), \cdots, B(t_n) - B(t_{n-1}) \tag{20}$$
are independent.
- **Normally distributed increments:** If $s, t \geq 0$, then
$$\mathbb{P}[B(t+s) - B(s) \in A] = \int_A \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t} dx. \tag{21}$$
- **Continuous trajectories:** With probability 1, $B_0 = 0$ and $t \mapsto B_t$ is continuous.

2 Properties of the Brownian Motion

The limiting object described above is called a *standard Brownian motion*. It will be our main object of study, and the main building block in many applications.

2.1 Covariance of Brownian Motion

Let $0 \leq s \leq t$, then

$$\mathbb{E}[B(s)B(t)] = \mathbb{E}[B(s)(B(t) - B(s) + B(s))] = \mathbb{E}[B(s)(B(t) - B(s))] + \mathbb{E}[B^2(s)]. \tag{22}$$

By independence of increments, we have that $\mathbb{E}[B(s)(B(t)-B(s))] = \mathbb{E}[B(s)]\mathbb{E}[B(t)-B(s)] = 0$. Moreover, from the fact that $B(s)$is a normally distributed random variable with expectation 0 and variance s, we have that $\mathbb{E}[B^2(s)] = s$. Bringing these results into the previous expression, we have that

$$\mathbb{E}[B(t)B(s)] = 0 + s = s. \tag{23}$$

Note that it is important to know how to compute the moments of a normal random variable.

2.2 Non-differentiability of Brownian Paths

In the following theorem we prove that the trajectories of Brownian motion are very special, in that they are non-differentiable at every point.

Theorem 2.1. With probability 1, Brownian paths are not differentiable at any point.

Proof. Let $W = \{\omega;\ \exists t \in (0,1)$, such that $B(t)$ is differentiable at $t\}$. We want to show that $\mathbb{P}[W] = 0$. We shall use the following lemma, whose proof is fairly straightforwards and will be left as an exercise.

> *Lemma. If $f : (0,1) \to \mathbb{R}$ is differentiable at $x \in (0,1)$, then there exists $C > 0$ and $\delta > 0$, such that $f(x) - f(s) \le C|x - s|$, for any $s \in [x - \delta, x + \delta]$.*

Now let $M_n(k) = \max_i \{|B((i+1)/n) - B(i/n)|;\ i = k-1, k, k+1\}$, and $M_n = \min_k \{M_n(k);\ k = 1, \cdots, n\}$. By the above lemma, for any $\omega \in W$, we can find $C > 0$ and $\delta > 0$, such that

$$|B(t) - B(s)| \le C|t - s|, \qquad \forall s \in [t - \delta, t + \delta]. \tag{24}$$

Choose n sufficiently large such that $n > \max\{t, 4/\delta\}$. We also choose k such that $(k-1)/n \le t \le k/n$. The choices of k and n imply that $|i/n - t| < \delta$ for $i = k-1, k, k+1, k+2$. Furthermore, we can see that

$$\left|B\left(\frac{i+1}{n}\right) - B\left(\frac{i}{n}\right)\right| \le \left|B\left(\frac{i+1}{n}\right) - B(t)\right| + \left|B\left(\frac{i}{n}\right) - B(t)\right| \le C\left|\frac{i+1}{n} - t\right| + C\left|\frac{i}{n} - t\right| < 2C\delta < \frac{8C}{n}. \tag{25}$$

Note that $B((i+1)/n) - B(i/n)$ are independent and normally distributed with mean 0 and variance $1/n$, so that

$$\mathbb{P}\left[|M_n| \le \frac{8C}{n}\right] \le n\mathbb{P}\left[|M_n(k)| \le \frac{8C}{n}\right] = n\left(\mathbb{P}\left[|z| \le \frac{8C}{\sqrt{n}}\right]\right)^3 \le n\left(\frac{16C}{\sqrt{2\pi n}}\right)^3 = \frac{16^3C^3}{\sqrt{2\pi n}}, \tag{26}$$

where z is the standard normal. Let $A_n := \{|M_n| \le 8C/n\}$, then the above inequality means that $\lim_{n\to\infty} \mathbb{P}[A_n] = 0$. By Fatou's lemma, we can conclude that

$$\mathbb{P}\left[\liminf_{n\to\infty} A_n\right] \le \liminf_{n\to\infty} \mathbb{P}[A_n] = \lim_{n\to\infty} \mathbb{P}[A_n] = 0. \tag{27}$$

Finally, we note that $W \subseteq \liminf_{n\to\infty} A_n$, so $\mathbb{P}[W] = 0$, and the proof is complete. □

2.3 Scaling Properties of the Brownian Motion

If we fix time t, then the distribution of Brownian motion $B(t)$ is normal with expectation 0 and variance t. In fact, for any fixed set of times $t_1 < t_2 < \cdots < t_n$, the random variables $B(t_1), B(t_2), \cdots, B(t_n)$ are jointly normal with mean 0 and covariance given by $\mathbb{E}[B(s)B(t)] = \min\{s, t\}$.

Theorem 2.2. If $B(0) = 0$, then for any $\lambda > 0$, the stochastic process $B(\lambda t)/\sqrt{\lambda}$, $t \ge 0$ is a Brownian motion.

Proof. First, note that $X(t) = B(\lambda t)/\sqrt{\lambda}$ is a Gaussian process, *i.e.*, for any set $t_1 < t_2 < \cdots < t_n$, the joint distribution of $X(t_1), X(t_2), \cdots, X(t_n)$ is a multivariate Gaussian distribution. This property is clearly inherited from Brownian motion properties. Since normal distribution is characterized by its mean and covariance, we have to check that the mean and covariance of the process $X(t)$ coincide with those of Brownian motion. Indeed,

$$\mathbb{E}[X(t)] = \mathbb{E}\left[\frac{B(\lambda t)}{\sqrt{\lambda}}\right] = \frac{\mathbb{E}[B(\lambda t)]}{\sqrt{\lambda}} = 0, \tag{28}$$

and for $s < t$, we have that

$$\mathbb{E}[X(s)X(t)] = \mathbb{E}\left[\frac{B(\lambda s)}{\sqrt{\lambda}} \cdot \frac{B(\lambda t)}{\sqrt{\lambda}}\right] = \frac{\mathbb{E}[B(\lambda s)B(\lambda t)]}{\lambda} = \frac{\min\{\lambda s, \lambda t\}}{\lambda} = s. \tag{29}$$

Therefore, we can conclude that $B(\lambda t)/\sqrt{\lambda}$ has the same distribution as a Brownian motion, and thus a Brownian motion. The proof is thus complete. □

Theorem 2.3. If $B(t)$ is a Brownian motion starting at 0, then so is the process defined by $X(0) = 0$ and $X(t) = tB(1/t)$ for $t > 0$.

Proof. Fix $t_1 < t_2 < \cdots < t_n$, then clearly $X(t_1), X(t_2), \cdots, X(t_n)$ has a multivariate Gaussian distribution. We just have to check that it has the same mean and covariance structure. First of all, we have that

$$\mathbb{E}[X(t)] = \mathbb{E}[tB(1/t)] = t\mathbb{E}[B(1/t)] = 0. \tag{30}$$

Moreover, for $0 < s < t$, we can deduce that

$$\mathbb{E}[X(s)X(t)] = \mathbb{E}[sB(1/s) \cdot tB(1/t)] = st\mathbb{E}[B(1/s)B(1/t)] = st \min\{1/s, 1/t\} = s. \tag{31}$$

Therefore, $X(t)$ is indeed a Brownian motion, so the proof is complete. □

2.4 Quadratic Variation of Brownian Motion

Last time we computed quadratic variation of scaled SRW and it turned out to be t. In the following theorem we prove that the quadratic variation of the Brownian motion is also t. Let me again emphasize that the paths of Brownian motion are unusual in that their quadratic variation is not zero. This makes stochastic calculus different from ordinary calculus. In fact, non-zero quadratic variation is the source of the *volatility* term in the *Black-Scholes equation*.

To see how Brownian motion is different from functions we are used to in ordinary calculus, consider a function $f : \mathbb{R} \to \mathbb{R}$ that is continuously differentiable and compute its quadratic variation up to time T. Let us introduce the norm of the partition $0 = t_0 < t_1 < \cdots < t_n = T$, such that

$$\|\Pi\| = \max_{0 \le j \le n-1} (t_{j+1} - t_j). \tag{32}$$

Then, its quadratic variation can be computed as

$$\langle f, f \rangle_T = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} (f(t_{j+1}) - f(t_j))^2. \tag{33}$$

We will use the mean value theorem which says that in each interval (x_1, x_2), there exists a point x_0, such that

$$f(x_2) - f(x_1) = f'(x_0)(x_2 - x_1). \tag{34}$$

Applying to each interval (t_j, t_{j+1}), we get that there is a point t_j^* in each such interval, such that

$$f(t_{j+1}) - f(t_j) = f'(t_j^*)(t_{j+1} - t_j). \tag{35}$$

Therefore, we can compute that

$$\begin{aligned} \sum_{j=0}^{n-1} (f(t_{j+1}) - f(t_j))^2 &= \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j)^2 \le \max_{0 \le j \le n-1} |t_{j+1} - t_j| \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j) \\ &= \|\Pi\| \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j) \to \|\Pi\| \int_0^T |f'(t)|^2 dt \to 0, \qquad \text{as } \|\Pi\| \to 0. \end{aligned} \tag{36}$$

Therefore, we have shown that continuously differentiable function has zero quadratic variation. For this reason we rarely consider quadratic variation in ordinary calculus.

Theorem 2.4. Let $B(t)$ be a Brownian motion. Then $\langle B, B \rangle_T = T$ almost surely.

Proof. Define

$$Q_\Pi = \sum_{j=0}^{n-1} (B(t_{j+1}) - B(t_j))^2. \tag{37}$$

We shall prove that $\mathbb{E}[Q_\Pi] \to T$ and $\mathrm{Var}[Q_\Pi] \to 0$ as $\|\Pi\| \to 0$. In other words, we shall prove the L^2 convergence of Q_Π, which is stronger than *almost sure* convergence. First, notice that

$$\mathbb{E}[(B(t_{j+1}) - B(t_j))^2] = \mathrm{Var}[B(t_{j+1}) - B(t_j)] = t_{j+1} - t_j. \tag{38}$$

□

Therefore, we can compute that

$$\mathbb{E}[Q_\Pi] = \sum_{j=0}^{n-1} \mathbb{E}[(B(t_{j+1}) - B(t_j))^2] = \sum_{j=0}^{n-1} (t_{j+1} - t_j) = T. \tag{39}$$

Now, in order to show that $\mathrm{Var}[Q_\Pi] \to 0$ as $\|\Pi\| \to 0$, note that

$$\begin{aligned}
\mathrm{Var}[(B(t_{j+1}) - B(t_j))^2] &= \mathbb{E}[((B(t_{j+1}) - B(t_j))^2 - (t_{j+1} - t_j))^2] \\
&= \mathbb{E}[(B(t_{j+1}) - B(t_j))^4 - 2(t_{j+1} - t_j)(B(t_{j+1}) - B(t_j))^2 + (t_{j+1} - t_j)^2] \\
&= \mathbb{E}[(B(t_{j+1}) - B(t_j))^4] - 2(t_{j+1} - t_j)\mathbb{E}[(B(t_{j+1}) - B(t_j))^2] + (t_{j+1} - t_j)^2 \\
&= 3(t_{j+1} - t_j)^2 - 2(t_{j+1} - t_j)^2 + (t_{j+1} - t_j)^2 = 2(t_{j+1} - t_j)^2.
\end{aligned} \tag{40}$$

In the previous deduction, we used the fourth moment of a normal random variable. Now, we can see that

$$\mathrm{Var}[Q_\Pi] = \sum_{j=0}^{n-1} \mathrm{Var}[(B(t_{j+1}) - B(t_j))^2] = \sum_{j=0}^{n-1} 2(t_{j+1} - t_j)^2 \le 2\,\|\Pi\| \sum_{j=0}^{n-1} (t_{j+1} - t_j) = 2\,\|\Pi\|\, T, \tag{41}$$

which clearly tends to 0 as $\|\Pi\| \to 0$. Therefore, we can conclude that $\langle B, B\rangle_T = \lim_{\|\Pi\|\to 0} Q_\Pi$ which has expectation T and variance 0, thus being T almost surely. Our proof is thus complete.

Remark 2.5. $\langle B, B\rangle_t = t$ can be informally written as

$$dB(t)dB(t) = dt. \tag{42}$$

Remark 2.6. In addition to computing the quadratic variation of Brownian motion, we can also compute the cross variation of $B(t)$ with t and the quadratic variation of t itself, which are

$$\lim_{\|\Pi\|\to 0} \sum_{j=0}^{n-1} (B(t_{j+1}) - B(t_j))(t_{j+1} - t_j) = 0, \qquad \lim_{\|\Pi\|\to 0} \sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 = 0. \tag{43}$$

These can be informally written as

$$dB(t)dt = 0, \qquad dtdt = 0. \tag{44}$$

4/10 Lecture

2.5 Volatility of Geometric Brownian Motion

Let α and σ be constants and define the *geometric Brownian motion* as

$$S(t) = S(0) \exp\left(\sigma B(t) + \left(\alpha - \frac{\sigma^2}{2} t\right)\right). \tag{45}$$

This is the asset-price model used in the Black-Scholes formula. We show how to use the quadratic variation of Brownian motion to identify the *volatility* σ from a path of the process. Let $0 \le T_1 \le T_2$ be given and suppose that we observe the geometric Brownian motion $S(t)$ on the time interval $T_1 \le t \le T_2$. We may then choose a partition of this interval $T_1 = t_0 < t_1 < \cdots < t_m = T_2$ and observe the "log-return" over each interval $[t_j, t_{j+1}]$, such that

$$\log \frac{S(t_{j+1})}{S(t_j)} = \sigma\,(B(t_{j+1}) - B(t_j)) + \left(\alpha - \frac{\sigma^2}{2}\right)(t_{j+1} - t_j). \tag{46}$$

The sum of the squares of the log-returns, sometimes called the *realized volatility*, is

$$\sum_{j=0}^{m-1}\left(\log\frac{S(t_{j+1})}{S(t_j)}\right)^2 = \sigma^2\sum_{j=0}^{m-1}\left(B(t_{j+1})-B(t_j)\right)^2 + \left(\alpha-\frac{\sigma^2}{2}\right)^2\sum_{j=0}^{m-1}(t_{j+1}-t_j)^2$$
$$+\,2\sigma\left(\alpha-\frac{\sigma^2}{2}\right)\sum_{j=0}^{m-1}\left(B(t_{j+1})-B(t_j)\right)(t_{j+1}-t_j)\to\sigma^2(T_2-T_1). \tag{47}$$

3 Construction of the Itô Integral

Consider an asset whose price per share is equal to X_t, $t\geq 0$. Also consider a portfolio that initially consists of Δ_0 shares. Now consider the following trading strategy. Keep the initial position Δ_0 up to time $t_1\geq t_0=0$ and then rebalance the portfolio by taking position Δ_1 in the asset. Keep it up to time $t_2\geq t_1$ and then rebalance the portfolio again by taking position Δ_2 in the asset. In general, we rebalance the portfolio at trading date t_i by taking position Δ_i in the asset and keeping it till the next trading date t_{i+1}. Then we are interested in the profit $I_T(\Delta)$ of the above trading strategy at time T. Clearly, we have that

$$I_T(\Delta)=\Delta_0(X(t_1)-X(t_0))+\Delta_1(X(t_2)-X(t_1))+\cdots+\Delta_n(X(t_n)-X(t_{n-1})), \tag{48}$$

and by an analogy with the Riemann integral, we write symbolically that

$$I_T(\Delta)=\int_0^T\Delta(t)dX(t), \tag{49}$$

where $\Delta(t)$ is a piecewise constant function which is equal to Δ_i on $[t_i,t_{i+1}]$.

3.1 Construction of the Stochastic Integral

We fix an interval $[S,T]$ and try to make sense of

$$\int_S^T f(t,w)dX_t(w), \tag{50}$$

where $f(t,w)$ is a random function and $dX_t(w)$ refers to the increments of stochastic process X_t. Before we proceed we make the following clarifications.

Remark 3.1. (1) We restrict our attention to functions f, such that for any fixed t, the random variable $f(t,w)$ is $\mathscr{F}_t$-measurable. To explain this restriction we come back to the previous example. Position Δ_i that we take in the asset of time t_i, $i\geq 1$ may depend on the price history $\mathscr{F}_t$ of the asset, but it must be independent of the future behavior of the process X_t.

(2) We restrict our consideration to the case where X_t is a Brownian motion, though the case of general stochastic process X_t is quite similar.

The problem we face when trying to assign meaning to the integral (50) is that Brownian motion paths cannot be differentiated with respect to time. If $X(t)$ is a differentiable function, then we can define

$$\int_S^T f(t,w)dX(t)=\int_S^T f(t,w)X'(t)dt, \tag{51}$$

where the right-hand side is an ordinary integral with respect to time. This approach unfortunately does not work for Brownian motion as we have proved that the trajectories of Brownian motion are not differentiable.

Just like in the definition of the usual Riemann integral $\int_S^T f(t)dt$, where $f(t)$ is a deterministic function, we start with a definition for a simple class of functions f and then extend by some approximation procedure. Assume that $\Pi=\{t_0,\cdots,t_n\}$ is a partition of $[S,T]$, *i.e.*, $S=t_0<t_1<\cdots<t_n=T$, and that $f(t,w)$ is constant in t on each subinterval $[t_j,t_{j+1})$. Such a process $f(t,w)$ is called a *simple process*. We start with defining integral (50) for

simple processes. Consider the interval $[t_0, t_1)$, on which $f(t,w) = e_1(w)$ is a random quantity, but independent of t. Therefore, it is natural to assume that

$$\int_{t_0}^{t_1} f(t,w)dB_t(w) = \int_0^1 e_1(w)dB_t(w) = e_1(w)(B(t_1) - B(t_0)). \tag{52}$$

Applying this procedure to intervals $[t_1, t_2), [t_2, t_3), \cdots, [t_{n-1}, t_n)$, we can obtain that

$$\int_S^T f(t,w)dB_t(w) = e_1(w)(B(t_1) - B(t_0)) + e_2(w)(B(t_2) - B(t_1)) + \cdots + e_n(w)(B(t_n) - B(t_{n-1})). \tag{53}$$

Now, to extend this definition of stochastic integral (50) for general processes $f(t,w)$, we approximate it with simple processes in a similar way that we approximate continuous functions by stepwise constant functions in the theory of Riemann integration. But without any further assumption on approximating functions $e_i(w)$, our definition of the integral leads to difficulties. Here is an example of what kind of difficulties we can expect. Consider

$$\int_0^T B_t dB_t. \tag{54}$$

Riemann integral is just a limit of Riemann sums, such that

$$\int_S^T f(t)dt \approx \sum_i f(t_i^*)(t_{i+1} - t_i), \tag{55}$$

where t_i^* is any point on the interval $[t_i, t_{i+1}]$. When the length of the longest interval in the partition tends to zero, we will obtain the correct limit. Note that it was not important what point t_i^* we take. For instance, it could be $t_i^* = t_i$ (left point approximation) or $t_i^* = t_{i+1}$ (right point approximation) or something else. Let us try to do something similar for integral (54). The left point approximation can be written as

$$I_1 \sim \sum_i B(t_i)(B(t_{i+1}) - B(t_i)), \tag{56}$$

and the right point approximation can be written as

$$I_2 \sim \sum_i B(t_{i+1})(B(t_{i+1}) - B(t_i)). \tag{57}$$

From the independence of increments of Brownian motion and the fact that $\mathbb{E}[B(t_{i+1}) - B(t_i)] = 0$ and $\mathbb{E}[B(t_1)] = 0$, we have that

$$\mathbb{E}[I_1] = \sum_i \mathbb{E}[B(t_i)(B(t_{i+1}) - B(t_i))] = \sum_i \mathbb{E}[B(t_i)]\mathbb{E}[B(t_{i+1}) - B(t_i)] = 0, \tag{58}$$

$$\mathbb{E}[I_2] = \sum_i \mathbb{E}[B(t_{i+1})(B(t_{i+1}) - B(t_i))] = \sum_i \big(\mathbb{E}[B^2(t_{i+1})] - \mathbb{E}[B(t_{i+1})B(t_i)]\big) = \sum_i (t_{i+1} - t_i) = T, \tag{59}$$

where in right point approximation, the result follows by observing that $\mathbb{E}[B^2(t_{i+1})] = t_{i+1}$ (quadratic variation) and $\mathbb{E}[B(t_{i+1})B(t_i)] = \min(t_{i+1}, t_i) = t_i$. Consequently, we can see that depending on the choice of the point t_i^* in the approximation, we can get very different results, unlike in Riemann integrals where such a choice does not matter. Now since the function $f(t,w)$ is $\mathscr{F}_t$-measurable, it is reasonable to choose the approximating simple function to be $\mathscr{F}_t$-measurable as well, so the left point approximation $t_i^* = t_i$ would be our only choice. This leads to the definition of the *Itô integrals*.

Remark 3.2. If for each $t \geq 0$, the random variable $f(t,w)$ is $\mathscr{F}_t$-measurable, then we say that the process $f(t,w)$ is $\mathscr{F}_t$*-adapted*. For instance, if $\mathscr{F}_t$ is a filtration of Brownian motion, then the process $f_t(t,w) = B(t/2)$ is $\mathscr{F}_t$-adapted, while $f_t(t,w) = B(2t)$ is not.

3.2 Properties of the Itô Integral for Simple Processes

The Itô integral (50) is defined as the gain from treading in the martingale B_t. A martingale has no tendency to rise or fall, and hence it is to be expected that

$$I_t(f) = \int_0^t f(t,w)dB_t \tag{60}$$

also has no tendency to rise or fall.

Theorem 3.3. Itô integral is a martingale.

Proof. It suffices to prove for simple processes, since the same conclusion for general stochastic processes can be obtained up to taking limits. Let $W(t)$ be a simple process, $0 \le s < t \le T$ be given, and Π be a partition. Assume that there exists partition points $t_l < t_k$, such that $s \in [t_l, t_{l+1})$ and $t \in [t_k, t_{k+1})$. Let $D_j = W(t_{j+1}) - W(t_j)$ for for $0 \le j \le k-1$ and $D_k = W(t) - W(t_k)$. Thus we split the sum as

$$I(t) = \sum_{j=0}^{l-1} \Delta_j D_j + \Delta_l D_l + \sum_{j=l+1}^{k-1} \Delta_j D_j + \Delta_k D_k. \tag{61}$$

We investigate the four summands respectively. For the first summand, since $t_l \le s$, $\Delta_j D_j$ is $\mathscr{F}_s$-measurable for all $0 \le j \le l-1$. Hence, we have that

$$\mathbb{E}[\Delta_j D_j | \mathscr{F}_s] = \Delta_j D_j. \tag{62}$$

For the second summand, we can compute that

$$\mathbb{E}[\Delta_l D_l | \mathscr{F}_s] = \Delta_l \mathbb{E}[W(t_{l+1}) | \mathscr{F}_s] - \Delta_l \mathbb{E}[W(t_l) | \mathscr{F}_s] = \Delta_l (W(s) - W(t_l)). \tag{63}$$

For the third summand, we can compute for each $l+1 \le j \le k-1$ that

$$\begin{aligned} \mathbb{E}[\Delta_j D_j | \mathscr{F}_s] &= \mathbb{E}[\mathbb{E}[\Delta_j D_j | \mathscr{F}_{t_j}] | \mathscr{F}_s] = \mathbb{E}[\Delta_j \mathbb{E}[W(t_{j+1}) | \mathscr{F}_{t_j}] - \Delta_j \mathbb{E}[W(t_j) | \mathscr{F}_{t_j}] | \mathscr{F}_s] \\ &= \mathbb{E}[\Delta_j (W(t_j) - W(t_j)) | \mathscr{F}_s] = 0. \end{aligned} \tag{64}$$

The fourth summand is similar to the third, which would vanish in expectation when conditioned on $\mathscr{F}_s$. Combining the previous results, we can then conclude that

$$\mathbb{E}[I(t) | \mathscr{F}_s] = \sum_{j=0}^{l-1} \Delta_j D_j + \Delta_l (W(s) - W(t_l)) = I(s), \tag{65}$$

which implies that $I(t)$ is a martingale, so the proof is complete. □

Now, since $I_t(f)$ is a martingale and $I_0 = 0$, we have that $\mathbb{E}[I_t(f)] = 0$ for all $t \ge 0$. It then follows that $\mathrm{Var}[I_t(f)] = \mathbb{E}[I_t^2(f)]$, which can be evaluated by the formula in the following theorem.

Theorem 3.4. The Itô integral satisfies that

$$\mathbb{E}[I_t^2(f)] = \mathbb{E}\left[\int_0^t f^2(s,w) ds\right]. \tag{66}$$

Proof. For the simplicity of notation we introduce $\Delta B_i = B(t_{i+1}) - B(t_i)$ and $e_i = e_i(w)$. Then by definition, we can write that

$$I_t(f) = \int_0^t f(s,w) dB_s = \sum_i e_i \Delta B_i. \tag{67}$$

Squaring on both sides, we will obtain that

$$I_t^2(f) = \sum_i e_i^2 (\Delta B_i)^2 + 2 \sum_{i<j} e_i e_j \Delta B_i \Delta B_j. \tag{68}$$

Now by taking expectation on both side, we can see that

$$\begin{aligned} \mathbb{E}[I_t^2(f)] &= \sum_i \mathbb{E}[e_i^2 (\Delta B_i)^2] + 2 \sum_{i<j} \mathbb{E}[e_i e_j \Delta B_i \Delta B_j] = \sum_i \mathbb{E}[e_i^2] \mathbb{E}[(\Delta B_i)^2] + 2 \sum_{i<j} \mathbb{E}[e_i e_j] \underbrace{\mathbb{E}[\Delta B_i]}_{=0} \underbrace{\mathbb{E}[\Delta B_j]}_{=0} \\ &= \sum_i \mathbb{E}[e_i^2] \mathrm{Var}[\Delta B_i] = \sum_i \mathbb{E}[e_i^2](t_{i+1} - t_i) = \mathbb{E}\left[\sum_i e_i^2 (t_{i+1} - t_i)\right] = \mathbb{E}\left[\int_0^t f^2(s,w) ds\right], \end{aligned} \tag{69}$$

as desired, so the proof is complete. □

Theorem 3.5. The quadratic variation of the stochastic integral

$$\left\langle \int_0^t f(t,w)dB_t, \int_0^t f(t,w)dB_t \right\rangle_T = \int_0^T f^2(t,w)dt = \sum_i e_i^2 \Delta t_i. \tag{70}$$

Proof. We take $\Pi = \{t_0, \cdots, t_n\}$ to be a partition of $[0,T]$, *i.e.*, $0 = t_0 < t_1 < \cdots < t_n = T$, then the quadratic variation can be computed as

$$\begin{aligned} Q_\Pi = \left\langle \int_0^t f(t,w)dB_t, \int_0^t f(t,w)dB_t \right\rangle_T &= \lim_{\|\Pi\|\to 0} \sum_{j=0}^{n-1} \left(\int_0^{t_{j+1}} f(t,w)dB_t - \int_0^{t_j} f(t,w)dB_t \right)^2 \\ &= \lim_{\|\Pi\|\to 0} \sum_{j=0}^{n-1} \left(e_j(w)(B_{t_{j+1}} - B_{t_j}) \right)^2. \end{aligned} \tag{71}$$

To this end, we consider the quantity

$$\begin{aligned} &\mathbb{E}\left[\left(\sum_{j=0}^{n-1} \left(e_j(w)(B_{t_{j+1}} - B_{t_j}) \right)^2 - \sum_{j=0}^{n-1} e_j(w)^2 (t_{j+1} - t_j) \right)^2 \right] \\ &= \mathbb{E}\left[\left(\sum_{j=0}^{n-1} \underbrace{e_j(w)}_{=:e_j}{}^2 \left(\underbrace{(B_{t_{j+1}} - B_{t_j})}_{=:\Delta B_j}{}^2 - \underbrace{(t_{j+1} - t_j)}_{=:\Delta t_j} \right) \right)^2 \right] \\ &= \sum_j \mathbb{E}\left[e_j^4 \left((\Delta B_j)^2 - \Delta t_j \right)^2 \right] + 2 \sum_{i<j} \mathbb{E}\left[e_i^2 e_j^2 \left((\Delta B_i)^2 - \Delta t_i \right) \left((\Delta B_j)^2 - \Delta t_j \right) \right] \\ &= \sum_j \mathbb{E}\left[e_j^4 \right] \mathbb{E}\left[\left((\Delta B_j)^2 - \Delta t_j \right)^2 \right] + 2 \sum_{i<j} \mathbb{E}\left[e_i^2 e_j^2 \right] \underbrace{\mathbb{E}\left[(\Delta B_i)^2 - \Delta t_i \right]}_{=0} \underbrace{\mathbb{E}\left[(\Delta B_j)^2 - \Delta t_j \right]}_{=0} \\ &= \sum_j \mathbb{E}\left[e_j^4 \right] \mathbb{E}\left[(\Delta B_j)^2 - \mathbb{E}\left[(\Delta B_j)^2 \right] \right] = \sum_j \mathbb{E}\left[e_j^4 \right] \mathrm{Var}\left[(\Delta B_j)^2 \right] = \sum_j \mathbb{E}\left[e_j^4 \right] \cdot 2 \left(\Delta t_j \right)^2 \le 2 \|\Pi\| \sum_j \mathbb{E}\left[e_j^4 \right] \Delta t_j. \end{aligned} \tag{72}$$

Now, if $\|\Pi\| \to 0$, we have that $\sum_j \mathbb{E}\left[e_j^4\right] \Delta t_j \to \int_0^T \mathbb{E}[f^4(t,w)]dt < \infty$, so that the expected value as computed above must vanish. As a result, the quadratic variation is shown to be

$$\sum_i e_i^2 \Delta t_i = \int_0^T f^2(t,w)dt, \tag{73}$$

as desired, and the proof is complete. □

3.3 Itô Integral for General Functions

We now introduce the class of general functions $f(t,w)$ for which is Itô integral will be defined.

Definition 3.6. Let $V = V(S,T)$ be the class of functions $f(t,w) : [0,\infty) \times \Omega \to \mathbb{R}$, such that $f(t,w)$ is $\mathscr{F}_t$-adapted and $\int_S^T f^2(t,w)dt < \infty$. The Itô integral is defined for $V(S,T)$.

We claim that each function $f \in V(S,T)$ can be approximated by a sequence $\{\phi_n\}_{n\ge 1}$ of simple functions (or equivalently, by a sequence of simple processes) in the sense that

$$\mathbb{E}\left[\int_S^T (f - \phi_n)^2 dt \right] \to 0, \qquad \text{as } n \to \infty. \tag{74}$$

The approximation is done in three steps.

(1) Approximate bounded continuous functions with simple functions. Let $g \in V$ be bounded, with every trajectory $g(\cdot, w)$ (w is fixed and t changes) being continuous. Then, there exists a sequence of simple functions $\phi_n \in V$, such that

$$\mathbb{E}\left[\int_S^T (g-\phi_n)^2 dt\right] \to 0, \qquad \text{as } n \to \infty. \tag{75}$$

(2) Approximate bounded functions with bounded continuous functions. Let $h \in V$ be bounded, then there exists a sequence of bounded continuous functions g_n, such that

$$\mathbb{E}\left[\int_S^T (h-g_n)^2 dt\right] \to 0, \qquad \text{as } n \to \infty. \tag{76}$$

(3) Approximate general functions with bounded functions. Let $f \in V$, then there exists a sequence of bounded functions h_n, such that

$$\mathbb{E}\left[\int_S^T (f-h_n)^2 dt\right] \to 0, \qquad \text{as } n \to \infty. \tag{77}$$

Putting together the three steps above, we get that for any function $f(t,w) \in V$, there exists a sequence of simple functions $\phi_n(t,w)$, such that (74) holds. We then define the Itô integral of the function $f(t,w)$ as

$$I(f) = \int_S^T f(t,w) dB_t = \lim_{n\to\infty} I(\phi_n). \tag{78}$$

The limit exists since by Theorem 3.4, we can deduce that

$$\mathbb{E}\left[(I(\phi_n) - I(\phi_m))^2\right] = \mathbb{E}\left[\int_S^T (\phi_n - \phi_m)^2 dt\right] \le \mathbb{E}\left[\int_S^T (f-\phi_n)^2 dt\right] + \mathbb{E}\left[\int_S^T (f-\phi_m)^2 dt\right] \to 0, \tag{79}$$

as $m, n \to \infty$. Thus, the sequence of random variables $\left\{\int_S^T \phi_n(t,w) dB_t\right\}$ forms a Cauchy sequence in $L_2(\Omega, \mathscr{F}, \mathbb{P})$. Since $L_2(\Omega, \mathscr{F}, \mathbb{P})$ is a complete space, there exists a limit of $I(\phi_n)$ as an element of it. This limit is by definition the Itô integral $I(f)$.

Example 3.7. Consider the Itô integral

$$\int_0^T B_t dB_t. \tag{80}$$

By definition, we can write that

$$\int_0^T B_t dB_t = \lim_{n\to\infty} \int_0^T \phi_n(t,w) dB_t, \tag{81}$$

where ϕ_n are $\mathscr{F}_t$-adapted simple functions such that

$$\mathbb{E}\left[\int_0^T (\phi_n - B_t)^2 dt\right] \to 0, \qquad \text{as } n \to \infty. \tag{82}$$

We now approximate $f(t,w) = B_t$ by partitioning $[0,T]$ into $[0,t_1], [t_1,t_2], \cdots, [t_{n-1}, T]$ and defining $\phi_n(t,w) = B(t_i)$ for $t \in [t_i, t_{i+1}]$. We check that

$$\begin{aligned}\mathbb{E}\left[\int_0^T (\phi_n - B_t)^2 dt\right] &= \mathbb{E}\left[\sum_i \int_{t_i}^{t_{i+1}} (\phi_n - B_t)^2 dt\right] = \mathbb{E}\left[\sum_i \int_{t_i}^{t_{i+1}} (B(t_i) - B_t)^2 dt\right] \\ &= \sum_i \int_{t_i}^{t_{i+1}} \mathbb{E}\left[(B(t_i) - B(t))^2\right] dt = \sum_i \int_{t_i}^{t_{i+1}} (t - t_i) dt = \sum_i \frac{(t_{i+1} - t_i)^2}{2}.\end{aligned} \tag{83}$$

By defining $M_n = \max_i(t_{i+1} - t_i)$, we can thus see that

$$\mathbb{E}\left[\int_0^T (\phi_n - B_t)^2 dt\right] = \sum_i \frac{(t_{i+1} - t_i)^2}{2} \leq \frac{M_n}{2} \sum_i (t_{i+1} - t_i) = \frac{M_n T}{2} \to 0. \tag{84}$$

Therefore, now it suffices to compute

$$\int_0^T \phi_n dB_t = \sum_i B(t_i) \underbrace{(B(t_{i+1}) - B(t_i))}_{=:\Delta B_i} = \sum_i B(t_i)\Delta B_i. \tag{85}$$

Note that we have the identity

$$\Delta B_i^2 = B^2(t_{i+1}) - B^2(t_i) = (\Delta B_i)^2 + 2B(t_{i+1})B(t_i) - 2B^2(t_i) = (\Delta B_i)^2 + 2B(t_i)\Delta B_i. \tag{86}$$

Summing both sides over i, we would then obtain that

$$\begin{aligned} B^2(T) &= \sum_i \Delta B_i^2 = \sum_i (\Delta B_i)^2 + 2\sum_i B(t_i)\Delta B_i = \sum_i (\Delta B_i)^2 + 2\int_0^T \phi_n dB_t \\ \implies \int_0^T \phi_n dB_t &= \frac{1}{2}\left(B^2(T) - \sum_i (\Delta B_i)^2\right) \to \frac{1}{2}\left(B^2(T) - T\right), \end{aligned} \tag{87}$$

where the limit is in the L_2 sense. Therefore, we can conclude that

$$\int_0^T B_t dB_t = \frac{B^2(T) - T}{2}. \tag{88}$$

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